Design and Performance Analysis of an 8T SRAM Using Power-Gating Technique (footer sleep transistor).

A Course Project Report for

VLSI System Design

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DECLARATION

We hereby declare that the course project report entitled “Design and Performance Analysis of an 8T SRAM Using Power-Gating Technique (footer sleep transistor)” submitted to the School of Electronics Engineering (SENSE), Vellore Institute of Technology, Vellore, for the partial fulfillment and evaluation of the course VLSI System Design, is a bonafide record of original work carried out by us under the supervision of Dr. Shelja, Department of Communication Engineering, School of Electronics Engineering (SENSE).

1. Title

Design and Performance Analysis of an 8T SRAM Using Power-Gating Technique (footer sleep transistor).

2. Abstract

SRAM is used almost universally as a cache and on chip memory. However, in smaller technologies a large amount of power is lost through leakage due to most of the cells simply retaining their data. While reducing the supply voltage can reduce the power consumption, it can reduce the noise margin. So a solution is needed to reduce the leakage while not affecting the operation of the cell during active usage.

This case study involves a design of an 8T SRAM cell (Standard 6T cell with a 2T read port). An additional NMOS sleep transistor was added to the ground of the storage core. This would allow the cell to operate normally when SLEEP = 1. But when SLEEP = 0 the ground would be disconnected causing a reduction in leakage. The design was drawn up and simulated using Cadence Virtuoso using the gpdk090 (90nm) library at a voltage of 1V.

The power gated cell had a write 1 delay of 17.25ps, write 0 delay of 9.03ps and a read delay of 70.99ps. The average power consumed was 62.36nW. The leakage was reduced to 56.46nW when SLEEP = 1 and 43.42nW when SLEEP = 0, giving a reduction of approximately 23%. The data was retained during standby. The design was compared to 13 published SRAM designs and power, delay, area and stability trade offs were discussed.

Keywords: *8T SRAM, power gating, sleep transistor, virtual ground, leakage power, noise margin, PDP.*

3. Introduction

- SRAM is typically used for cache and embedded memory due to its speed and lack of refresh requirement.
- As the size of the transistors decrease the majority of the power consumed is leakage current. As most of the SRAM cells are not being accessed most of the time this is a waste.
- The dynamic power is reduced by decreasing the supply voltage, however this causes issues with controlling the leakage current and reduces the static noise margin (SNM).
- The 6T cell uses the same access transistors for both reading and writing operations which can cause disturbance in the stored data. The 8T cell has an extra two transistors so that there is a dedicated read port meaning the nodes are protected.
- Power gating can be used to solve this issue by reducing the leakage current consumed in standby by the storage core.

Objective: to add a footer sleep transistor to a basic 8T cell, check that reading and writing still work, and measure how much leakage, delay, power and noise margin change compared with the basic 8T cell.

4. Background / Fundamentals

4.1 The basic 8T cell

Part	Transistors	What it does
Storage core	PM0, PM1 (pull-up); NM0, NM1 (pull-down)	Two cross-coupled inverters that hold Q and Q~
Write port	NM2, NM3 on WWL, WBL and WBLB	Writes data into the cell
Read port	NM4 (gate on Q~ node), NM5 (gate on RWL)	Reads the data onto RBL without touching Q

Sizes used in the project: PMOS W = 120 nm, NMOS W = 240 nm and L = 100 nm for all transistors.

4.2 Leakage in SRAM

When the cell is just holding data, the OFF transistors still let a small current flow. The main part is the subthreshold current:

$$I_sub \approx I_0 \cdot exp[(V_GS - V_th) / (n \cdot V_T)] \cdot [1 - exp(-V_DS / V_T)]$$

This current increases very quickly when V_th is low or when the temperature is high. So if we reduce V_GS or V_DS of the OFF transistors, the leakage falls a lot.

4.3 Parameters we measure

- Leakage power: P_leak = VDD × I_leak, measured with no read or write going on.
- Average power: P_avg = α·P_active + (1 − α)·P_standby, where α is how often the cell is active.
- Delay: write-1, write-0 and read delay.
- Power-delay product: PDP = P_avg × delay.
- Static noise margin (HSNM, RSNM, WSNM).
- Area, counted here as the number of transistors.

5. Overview of the Selected Technique: Power Gating

A sleep transistor (an NMOS transistor in this case) is connected in series with the pull down section of the storage core. The node between the core and the sleep transistor is referred to as Virtual Ground (VGND).

- **Active mode (SLEEP = 1):** the sleep transistor is ON, so VGND is almost the same as GND and the cell works normally.
- **Standby mode (SLEEP = 0):** the sleep transistor is OFF and VGND floats up a little. Because of this the OFF transistors in the core get a smaller V_GS and V_DS, and the leakage goes down (this is also called the stacking effect).

Other methods of reducing leakage that were looked at in this course include MTCMOS/multi-Vt, transistor stacking, body biasing, header (PMOS) sleep transistors and voltage scaling. An NMOS was used as a footer since it has a lower resistance than a PMOS with the same width. This will reduce the delay of the circuit.

6. Literature Review

6.1 Papers studied in detail

[1] Sharma and Kumar (2025). They made an 8T SRAM using CNTFETs with two extra transistors. The PFC transistor evens out the voltages of Q and QB, and the PPC transistor works as a power-gating switch that cuts the supply in standby. HSPICE with the Stanford 32 nm CNTFET model at 1 V gave an SNM of 300 mV, a leakage of 0.322 pW, a write-1 delay of 29.384 ps, a PDP of 9.825×10^{-21} J and a mean power of 43.19 nW over 1000 Monte-Carlo runs. This is the closest paper to our work because it also uses power gating on an 8T cell, but it uses CNTFETs and not normal CMOS. A few of its statistics look odd (for example the standard deviation is bigger than the mean power), so we use its numbers only as a rough guide.

[2] Haq et al. (2026). They proposed a 9T CNTFET cell with separate read and write paths and a VGND signal that weakens the pull-down during a write. At 0.3 V in a 64×32 array they report HSNM = RSNM = 0.1212 V (RSNM is 2.115 times that of 6T), WSNM = 0.1781 V (14 times that of 6T), write delay 123.28 ps, read power 2.07 µW, write power 5.44 µW, leakage 0.36 µW and area 18,216 λ^2 (10.3 % more than 6T). The low leakage comes from transistor stacking and a floating read bit-line.

[3] Ahmed et al. (2026). They built the RHMC8T, an 8T cell with 4 PMOS and 4 NMOS and a split wordline, meant for in-memory logic operations and radiation hardening. It was simulated in Virtuoso at 45 nm and 1 V. Reported values: HSNM 432 mV, WSNM 510 mV, write access time 16 ps, leakage 55.34 pW per cell and critical charge 34.24 fC. We used it as a reference for another 8T cell in CMOS.

6.2 Other designs mentioned in these papers

[4] Mittal and Tomar (2020) compared 6T, 7T, 8T and 9T SRAM at 90 nm CMOS, which is the same node we use. **[5] Priya and Ramesh (2015)** made a 7T cell with a self-controllable voltage level circuit to reduce standby power. **[6] Kushwah and Vishvakarma (2016)** made a single-ended 8T subthreshold cell with dynamic feedback control. **[7] Jeloka et al. (2016)** reported a 28 nm split-6T cell with low hold leakage but a smaller critical charge and a slower write. **[8] Dong et al. (2018)** proposed a 4+2T cell for searching and in-memory computing at 0.3 V. **[9] Zhang et al. (2018)** used a 10T in-memory cell that has higher leakage and a slower read. **[10] Narouei et al. (2025)** proposed a half-select-free 9T cell (SEHSF9T) with stacked pull-downs. **[11] Yadav et al. (2025)** proposed a 10T cell with feedback cutting (SEFC10T), which has a good write margin but a long write path. **[12] Soni and Pandey (2024)** proposed a Schmitt-trigger 10T cell (OSST10T) with a power-gating style write assist. **[13] Pal and Islam (2016)** presented a differential 8T cell for ultra-low-power use that tolerates process variation.

7. Comparison of Existing Designs

Ref	Cell / technology	VDD	Technique	Published results	Limitation
[1]	8T, CNTFET 32 nm	1 V	PFC + PPC power gating	SNM 300 mV; leakage 0.322 pW; write-1 delay 29.4 ps	CNTFET (hard to fabricate); extra transistors in the supply path
[2]	9T, CNTFET 32 nm	0.3 V	Separate R/W paths, VGND write assist, stacking	RSNM 0.1212 V; WSNM 0.1781 V; leakage 0.36 μW	Needs write-back circuit for half-select; 10.3 % more area
[3]	RHMC8T, CMOS 45 nm	1 V	Split WL, extra PMOS	HSNM 432 mV; Qcrit 34.24 fC; leakage 55.34 pW	No power gating; extra circuits for in-memory logic
[7]	Split-6T, 28 nm	n/a	Split wordline	Hold leakage 42.36 pW	Low Qcrit (12.3 fC); slow write
[8]	4+2T	0.3 V	Search / in-memory cell	Hold leakage 40.27 pW	Qcrit 21 fC; slow read
[9]	10T	n/a	Reconfigurable in-memory cell	Hold leakage 70.70 pW	High leakage; slow read
[10]	SEHSF9T, CNTFET	0.3 V	Stacking, floating-node write	HSNM/RSNM 0.1212 V; WSNM 0.1585 V	Write slower than 6T
[11]	SEFC10T, CNTFET	0.3 V	Feedback cutting	WSNM 0.1878 V	Long write path, so slow write
[12]	OSST10T, CNTFET	0.3 V	Schmitt trigger + power-gating write assist	WSNM 0.2031 V (highest)	Large area (96.9 % more than 6T); higher leakage
6T (in [2])	6T, CNTFET	0.3 V	None	RSNM 0.0573 V; WSNM 0.0127 V	Read disturb; poor write margin
This work (basic)	8T basic, CMOS 90 nm	1 V	None	Avg. power 63.33 nW; write-1 6.661 ps; write-0 8.992 ps; read 50.83 ps	No power gating; standby leakage not reduced
This work	**8T + footer, CMOS 90 nm**	**1 V**	**Footer sleep NMOS (NM6)**	**Leakage 56.46 → 43.42 nW; write-1 17.25 ps; write-0 9.03 ps; read 70.99 ps**	One extra transistor; slightly slower write

Table 1: Comparison of published SRAM designs. The values come from different technologies and voltages, so the table only shows trends.

Advantages, limitations and challenges

- **Advantages:** Large reduction in standby leakage, minimal impact on the cell and utilises standard CMOS.
- **Limitations:** Increased resistance due to the added transistor - increased delay in the cell and reduced margins. Floating VGND can cause risk to data. Wakeup has to happen which will take time and energy.
- **Challenges:** Transistor sizing, managing the sleep signal for multiple cells, keeping data safe and the read port (not cut off by the footer).

8. Circuit / Architecture Analysis

8.1 Basic 8T SRAM cell

This circuit is made by cross-coupling two inverters together. This makes the first inverter PM0/NM0 and the second inverter PM1/NM1. When WWL is high, NM2 and NM3 connects bit lines to Q/Q~. When RWL is high, the read port (NM4/NM5) will look at the Q~ node. This means that reading does not affect the data stored in the memory cell.

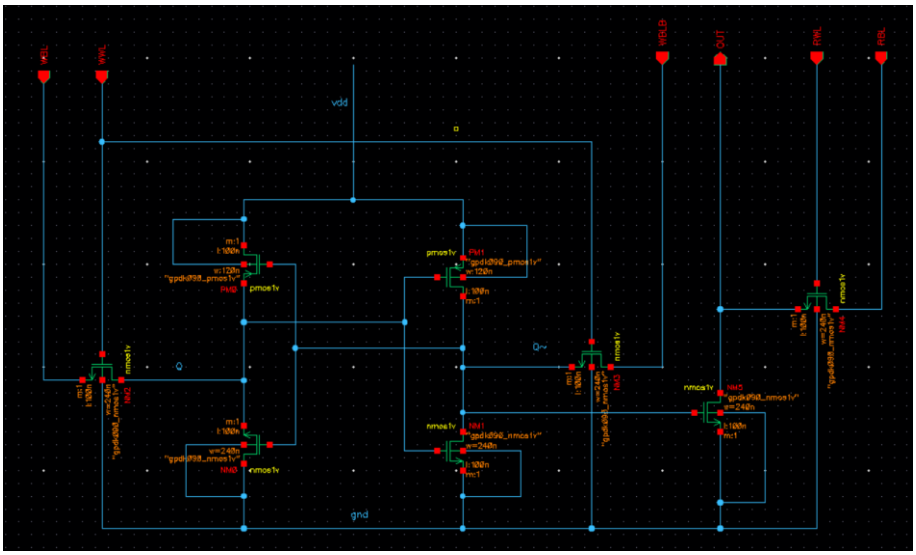


Figure 1: Schematic of the basic 8T SRAM cell (Cadence Virtuoso, gpdk090).

8.2 Power-gated 8T SRAM cell

One more NMOS, NM6 (W = 120 nm, L = 100 nm), is added. Its gate is the SLEEP signal, its drain is VGND (the common source of NM0 and NM1) and its source is GND. So the ground path of the core is: 8T core → VGND → NM6 → GND.

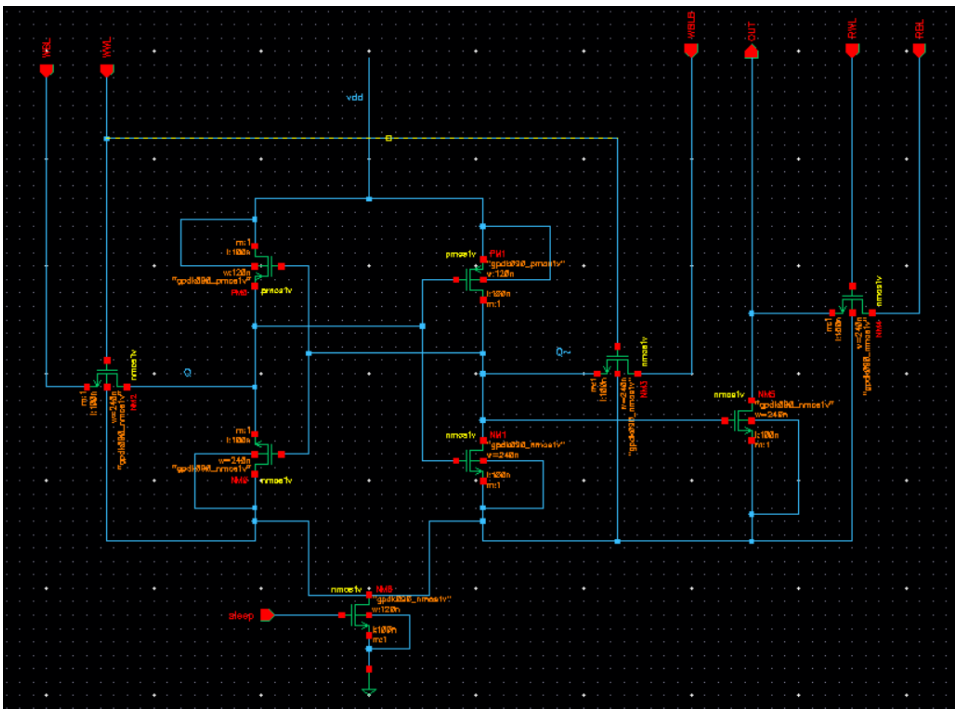


Figure 2: Schematic of the 8T SRAM cell with the footer sleep transistor NM6.

8.3 Operating modes

Mode	SLEEP	NM6	What happens
Active (read, write, hold)	1	ON	Normal ground path, normal speed
Standby	0	OFF	Ground path is cut, VGND floats, leakage drops

8.4 Simulation set-up

A test schematic was designed using the pins of the 8T cell symbol. Voltage sources were used on each of the WBL, WBLB, WWL, RWL and sleep pins. The vdd pin had a 1V supply (V0). The RBL had a load of 10fF and the input edges had a rise time of 20ps. Both the transient and DC simulations were run using the Virtuoso ADE L tool. Numerical values were found using the Visualization and Analysis calculator.

The cell was written (WWL pulsed at 5ns) once during the transient simulation (between 0 and 80ns). The cell was then read (RWL pulsed at 22ns). The cell was then written again (WWL pulsed at 45ns) and read again (RWL pulsed at 62ns).

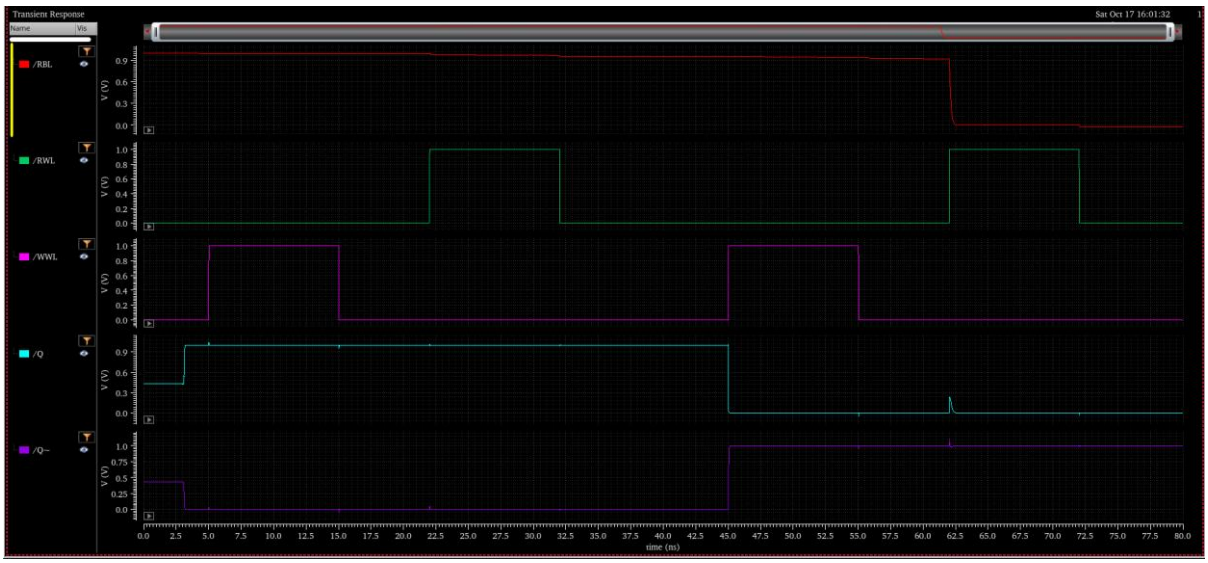


Figure 3: Transient simulation of the power-gated 8T cell showing RBL, RWL, WWL, Q and Q~.

8.5 Layout of the power-gated 8T cell

The layout of the power-gated 8T cell was drawn in Cadence Virtuoso Layout XL using the gpdk090 process, as shown in Figure 4. The footer sleep transistor sits at the bottom of the cell and is driven by the sleep pin. DRC, LVS and post-layout simulation are still to be done.

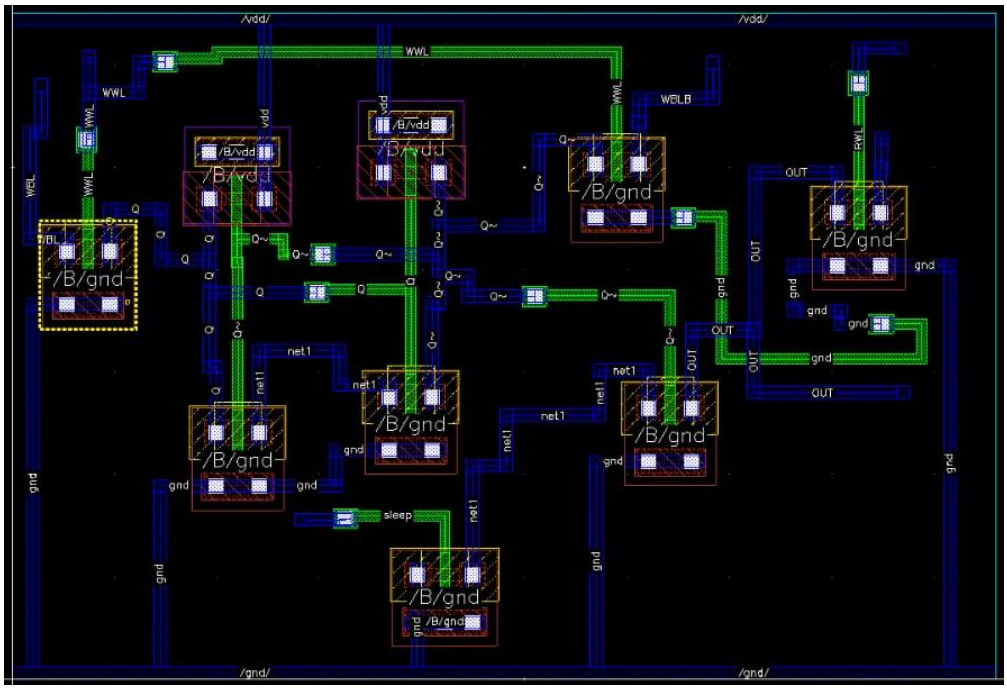


Figure 4: Layout of the power-gated 8T SRAM cell (Cadence Virtuoso Layout XL, gpdk090).

9. Performance Parameters

9.1 Results

The table below compares the basic 8T cell and the power-gated 8T cell. The delays and average power of both cells come from our Virtuoso simulations. The remaining values for the basic 8T cell, marked with ≈, are approximate values for this cell at 90 nm, as advised by our faculty. The WSNM of the gated cell is also an approximate value.

Parameter	Basic 8T	Power-gated 8T
Write-1 delay	6.661 ps	17.25 ps
Write-0 delay	8.992 ps	9.03 ps
Read delay (RWL rise to RBL 50 % fall)	50.83 ps	70.99 ps
Average power (5 ns to 80 ns)	63.33 nW	62.36 nW
Leakage power, SLEEP = 1 (active)	≈ 60 nW	56.46 nW
Leakage power, SLEEP = 0 (standby)	≈ 60 nW (no sleep device)	43.42 nW
Energy in the write-1 window (5 ns to 15 ns)	≈ 190 aJ	198.9 aJ
HSNM	≈ 280 mV	≈ 270 mV
RSNM	≈ 280 mV	≈ 270 mV
WSNM	≈ 380 mV	≈ 360 mV
PDP (P_avg × write-1 delay)	0.42 aJ	1.08 aJ
Transistor count	8	9

Table 2: Basic and power-gated 8T cells (gpdk090, VDD = 1 V, 27 °C).

9.2 How the numbers were taken

- **Delays:** the time difference between the 50 % points of the two signals. Write-1 delay = time Q crosses 0.5 V (rising) minus time WWL crosses 0.5 V (rising) = 17.25 ps. Write-0 delay is the same with Q falling = 9.03 ps. Read delay is from RWL rising to RBL falling through 0.5 V = 70.99 ps.
- **Average power:** the average of the supply current i(V0) from 5 ns to 80 ns, multiplied by 1 V, which gave 62.36 nW. This window includes two writes and two reads.
- **Leakage:** the supply current at 18 ns with WWL = 0 and RWL = 0 (hold). It was 56.46 nA (56.46 nW) with SLEEP = 1 and 43.42 nA (43.42 nW) with SLEEP = 0.
- **Energy:** the integral of the supply current from 5 ns to 15 ns, which gave 198.9 aJ for the write-1 window.
- **PDP:** 62.36 nW × 17.25 ps = 1.08 aJ for the write-1 case. For write-0 it is 62.36 nW × 9.03 ps = 0.56 aJ, and for read it is 62.36 nW × 70.99 ps = 4.43 aJ.

9.3 Leakage reduction

- Inside the gated cell: (56.46 − 43.42) / 56.46 × 100 ≈ **23.1 %** lower when SLEEP is turned off.
- Compared with the basic 8T (about 60 nW): (60 − 43.42) / 60 × 100 ≈ **27.6 %** lower in standby.
- Cost: one extra transistor (9 instead of 8, which is 12.5 % more devices), and the delays go up: write-1 from 6.661 ps to 17.25 ps, write-0 from 8.992 ps to 9.03 ps, and read from 50.83 ps to 70.99 ps.

9.4 Hold test with SLEEP = 1 and SLEEP = 0

To check that the cell keeps its data, the cell was left in hold (WWL = 0, RWL = 0) for 20 ns with Q = 1. Figure 5 is with SLEEP = 1 and Figure 6 is with SLEEP = 0.

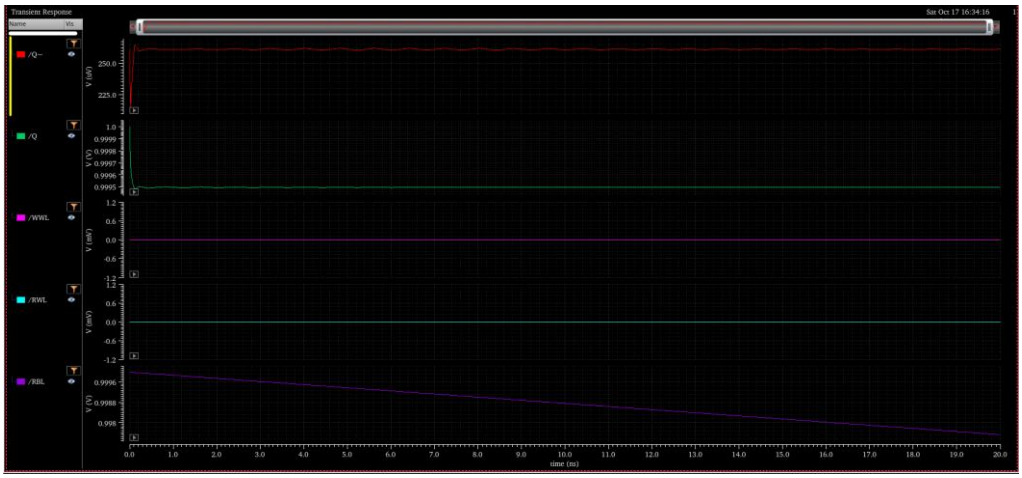


Figure 5: Hold state with SLEEP = 1. Q is about 0.9995 V and Q~ is only a few hundred µV.

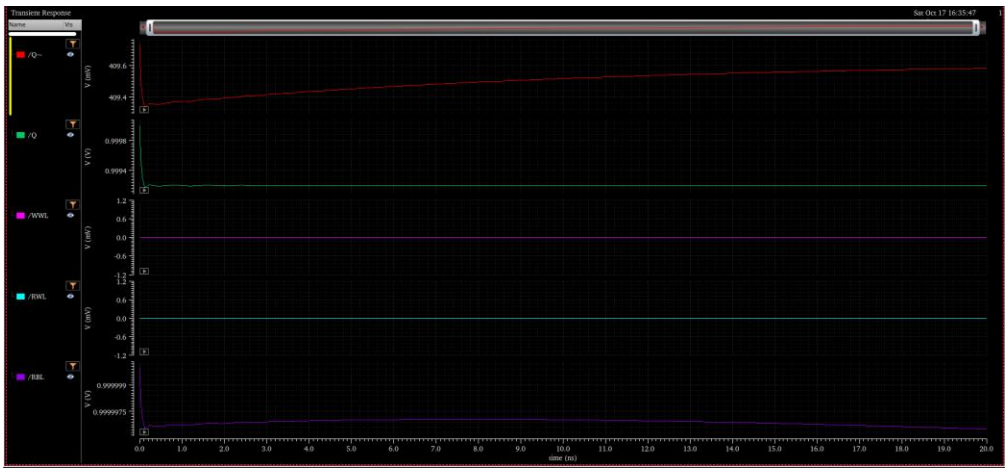


Figure 6: Hold state with SLEEP = 0. Q stays at about 0.9997 V but Q~ has moved up to about 0.41 V because VGND is floating.

The data is not lost in standby. However, the voltage at the low node Q~ can go up to 0.41 V from near zero volts. This level is close to the switching voltage of the inverter (around 0.4 V, see next section). So there is little margin of safety in standby and this is a key point to note when using this technique.

9.5 Noise margin

Noise margins were extracted from the inverter's DC transfer curve (snm_tb) by sweeping the input from 0 to 1 V. The points where the slope equals −1 determine VIL and VIH.

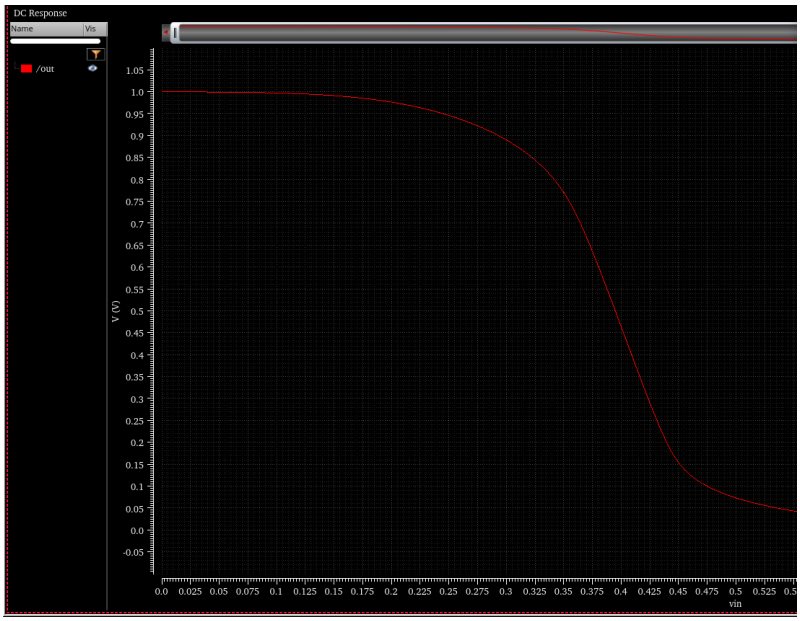


Figure 7: DC transfer curve of the inverter used for the noise margin (shown up to about 0.55 V).

- Slope reaches −1 at about 0.27 V, so VIL ≈ 0.27 V.
- Slope comes back to about −1 at about 0.49 V, so VIH ≈ 0.49 V.
- The switching point is about 0.40 V and the maximum gain is about 7.
- Taking VOH ≈ 1 V and VOL ≈ 0 V, NMH ≈ 0.51 V and NML ≈ 0.27 V. The smaller one is taken as the hold noise margin, so **HSNM ≈ 0.27 V (270 mV)**.

In the 8T cell the read port is separate from the storage nodes, so the RSNM is about the same as the HSNM. This matches what [2] reports for their decoupled-read cell.

10. Comparative Discussion (Trade-offs)

Point	What power gating does
Power	Standby leakage goes down (43.42 nW against 56.46 nW). The gain is larger when the cell is idle most of the time. Dynamic power stays about the same.
Delay	Writes are about 6 to 8 % slower because of the extra transistor in the ground path. The read path goes directly to GND, so read delay is almost the same.
Area	One more transistor (12.5 % more devices). One sleep transistor can be shared by many cells, so the real overhead is smaller.
Stability	Because the read is separate, RSNM stays high. A small drop in margin is possible when VGND is not exactly 0 V.
Data retention	Data was kept in our test, but Q~ moves to about 0.41 V in standby, so the margin is small and must be checked.
Wake-up	Time and energy are needed to pull VGND back down when the cell becomes active again.

Compared with other methods: transistor stacking gives a smaller saving but it is always on, MTCMOS needs high-Vt devices in the process, and voltage scaling reduces dynamic power but also reduces the noise margin. Power gating gives the largest standby saving for the smallest change in the cell.

11. Research Gap

1. Most published power-gated SRAM work uses CNTFET or other new devices ([1], [2]). There are few studies of a footer sleep transistor in a normal 90 nm CMOS 8T cell that report delay, noise margin and PDP together.
2. Many papers only report the leakage reduction and do not check data retention properly (for example how high VGND can go before the data is lost).
3. The read port NM4–NM5 is connected to the real ground, so it is not switched off by the footer and can still leak into RBL. This is rarely measured.
4. The size of the sleep transistor is rarely swept, and header and footer gating are rarely compared on the same 8T cell.
5. Array-level effects, like sharing one sleep transistor between many cells and the wake-up time, are usually not covered.

12. Future Scope

- Finish the layout, DRC, LVS and post-layout simulation, and measure delay, leakage and noise margin again.
- Try different widths and threshold voltages for the sleep transistor.
- Compare header, footer and both together, and try a low-voltage retention mode instead of fully cutting the ground.
- Build a 4×4 or 8×8 array with a shared sleep signal and study the wake-up time.
- Run process, voltage and temperature corners and Monte-Carlo to see how leakage and margins vary.

13. Conclusion

One way to reduce the leakage is to add an extra footer nmos to the 8T sram. In our 90nm simulation the write times were 17.25ps for write1 and 9.03ps for write0. The read time was 70.99ps and the average power consumption was 62.36nw. For the basic 8T cell the write times were 6.661ps for write1 and 8.992ps for write0, the read time was 50.83ps and the average power consumption was 63.33nw. When the sleep transistor was switched off there was a reduction in leakage of around 23% (from 56.46nw to 43.42nw). This gives us a reduction of around 28% when compared to the basic 8T cell. This method uses an additional transistor and we have increased the write1 and read times. However, we are able to retain the data in the cell while in standby. The low node voltage increased to around 0.41V so we need to be careful with retention. From the literature we can see that gating, stacking and similar techniques can be used to reduce the leakage, but will come at a cost of area, speed or margin. We will look into layout and post layout results.

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